

ResortHub™ Platform

Complete Technical & Functional Requirements Specification Document

DOCUMENT STATUS	ARCHITECTURE	SCOPE	DATE & TIME
Approved & Implemented	Flutter Web/Mobile + Client SaaS	5 Core User Roles & 8 Modules	September 2026

1. Executive Summary & Product Vision

ResortHub™ is an all-in-one multi-tenant hotel, resort, and farmhouse booking and hospitality management software suite. The platform bridges customer self-service exploration with complex back-office property operations, staff payroll, shift scheduling, real-time availability holds, food & beverage fulfillment, and multi-tier SaaS platform governance.

The frontend is built with Flutter to deliver a consistent, 60fps responsive user experience across Mobile (iOS/Android) and Web viewports. The system integrates a **Luxury Mint & Emerald Design System** with full reactive **Light and Dark Mode** capabilities.

Core Value Proposition

ResortHub eliminates fragmented hotel management by integrating customer reservations, double-booking exclusion constraints, express QR arrival passes, staff task dispatching, dining orders, accounting profit ledgers, and SaaS multi-resort onboarding into a unified single-source system.

2. User Roles & Permission Matrix

The platform enforces strict role-based access control (RBAC). Each persona accesses a dedicated, role-tailored interface:

Role	Target Persona	Permissions & Core Access	Primary Route
Customer / Guest	Travelers, families, and vacationers	Browse resorts, filter live amenities, view rooms, book stays, select advance/full pay, receive WhatsApp/Gmail passes, view QR badges.	/customer
Operations Incharge	Resort Supervisor / Property Incharge	Manage operational staff, assign housekeeping/maintenance tasks, log work schedules, record payroll disbursements, fulfill food orders.	/incharge
Resort Admin	General Manager / Property Owner	Manage room units, configure rate rules, block dates for maintenance, supervise incharge, broadcast promo SMS, view property financials.	/admin
Accountant	Financial Auditor / Controller	Inspect revenue summaries, reconcile food & stay transactions, review ledger settlements, audit staff payroll disbursements.	/accountant
Super Admin	Platform Co-Owner / SaaS Master	Manage platform-wide resort catalog (21 resorts), configure subscription tiers (Free, Basic, Super, Premium), onboard new resorts, assign admins.	/super-admin

3. Functional Requirements — Customer Discovery & Booking

3.1 Screen 1: Home / Discover (Luxury Mint Design)

- **Personalized Greeting Header:** Displays guest greeting (Hi, Rahul Verma 🙌) with real-time notification indicator.
- **Search Capsule:** Search capsule with interactive inputs for destination, date ranges, and guests.
- **"Escape The Ordinary" Banner:** Prominent hero showcase with curved border radius and high-contrast "Explore Now" trigger.
- **Popular Destination Chips:** Horizontal scrollable filter chips including *All Locations, Goa, Hyderabad, Munnar, Udaipur, Shimla, Coorg, Jaipur, Ooty, Rishikesh*.
- **Featured Luxury Offers Carousel:** Horizontal resort showcase cards displaying imagery, star rating badge, location, and emerald price tag (₹.../night).
- **Interactive Amenities Bar:** Direct toggles for top amenities on the discovery screen.

3.2 Screen 2: Search Results & Filter Pipeline

- **Dynamic Filter Header:** Shows location dropdown, interactive subtitle (guest count & date selection), and GPS location detector.
- **Live Amenity Filter Toggles:** Multi-select filter chips allowing guests to filter catalog by: **Free Wi-Fi, Swimming Pool, Luxury Spa, Restaurant, Free Parking, Beach Access, Lake View, Bonfire, Air Conditioning, Hot Water**.
- **Search Result Cards:** Full-width cards displaying:
 - High-resolution photo gallery with counter badge (e.g. 1/12).
 - Resort title, star rating, verified review count, and performance label (*Superb / Fabulous*).
 - **Interactive Location Line:** Displays exact city/state and incorporates a direct hyperlink that launches **Google Maps** coordinates in a new tab.
 - Cancellation and breakfast policy tags (e.g. *Free cancellation available, Breakfast included*).
 - Primary action button: **"View Rooms"** leading directly to Screen 3.

3.3 Screen 3: Resort & Room Details

- **Hero Image Slider:** Full-width interactive PageView gallery with circular back and favorite triggers.
- **Property Overview:** Capacity badge (2 Adults • 1 Child), star rating, and Google Maps location button.
- **4 Core Amenity Cards:** Highlighted visual tiles for **Wi-Fi, Breakfast, Free Parking, Swimming Pool**.
- **Property Amenities Tag Cloud:** Complete listing of all certified amenities with checkmark icons.
- **Available Room Units:** Room cards showing room name, features, nightly rate, and direct "Book" triggers.
- **Sticky Bottom Booking Card:** Floating reservation card displaying member exclusive badges, nightly rate, and a primary **"Book Room"** button.

4. Reservation, Split-Payment & Check-in QR Engine

Conflict-Free Date Availability

Prevents overlapping stays on any room unit using real-time availability verification. Units reserved by other guests show a red blocked banner and disable submission.

35% Advance Payment Split

Enables guests to choose between paying a **35% Advance Hold** or paying the **Full 100% Stay Amount** upfront, lowering booking drop-offs.

4.1 Reservation & Payment Flow Specification

1. **Guest Input:** Guest verifies full name, email, phone number, and guest count.
2. **Date Range Picker:** Interactive calendar with theme-adaptive styling and automated night calculation.
3. **Amount Breakdown:** Automatically calculates Total Stay Amount and Advance Required (35%).
4. **Payment Execution:** Demo payment gateway processes transactions with idempotency key protection.
5. **Confirmation & QR Code:** On success, reservation is committed to the database and generates a unique Express Check-in QR code containing encrypted reservation metadata.
6. **Dual Notification Outbox:** Generates real-time confirmation cards simulating:
 - **WhatsApp Confirmation:** Formatted welcome message, reservation ID, check-in dates, and check-in QR code.
 - **Gmail Confirmation:** Formal confirmation email with room summary, financial breakdown, and guest instructions.

4.2 "My Bookings" & Digital Pass Verification

- List of all active, upcoming, and past stays.
- Visual digital boarding pass with resort name, room category, arrival/departure dates, and payment status.
- Dynamic QR code display readable by front-desk staff for contactless arrival verification.

5. Back-Office Operations, Staff & Accounting Portals

5.1 Operations Incharge Portal (/incharge)

- **Staff Directory:** Add, activate, and manage resort ground staff (cleaners, chefs, maintenance, front desk).
- **Operational Tasks:** Assign priority tasks (e.g. *Pool Cleaning*, *Villa Sanitization*, *AC Repair*) with estimated completion times. Send real-time simulated SMS notifications to staff upon assignment.
- **Work Schedules:** Visual weekly shifts (Morning, Evening, Night) with assigned staff headcounts.

- **Staff Payroll Disbursements:** Pay monthly salaries with auto-generated receipt references and balance updates.
- **Food & Dining Fulfillment:** Track kitchen orders from guest rooms, update order statuses (*Preparing, Out for Delivery, Delivered*), and calculate revenue.

5.2 Resort Admin / Manager Portal (/admin)

- **Overview Dashboard:** Real-time KPIs for Today's Revenue, Monthly Revenue, Occupancy Rate, Arrivals, and Cancellations.
- **Room Inventory & Rates:** Add/edit room units, nightly base rates, and weekend rate rules.
- **Date Blocking:** Block date intervals with custom administrative reasons (e.g., *Private Wedding Buyout, Annual Maintenance*).
- **Incharge Supervision:** Assign and reassign Operations Incharge per property.
- **Promotional Messaging:** Broadcast promotional discount SMS alerts to previous guests.

5.3 Financial Accountant Portal (/accountant)

- **Dual Scoping:** Supports property-specific accountants as well as Platform Chief Accountants.
- **Revenue Ledgers:** Reconcile room bookings, advance payments, and remaining balance collections.
- **Dining Sales Audit:** Track F&B revenue against stay revenue.
- **Expense Tracking:** Review operational expenditure, maintenance costs, and payroll disbursements.

6. Super Admin / Co-Owner Multi-Tenant SaaS Portal

The Super Admin portal (/super-admin) provides high-level control over the entire SaaS platform, which currently manages **21 subscribed resorts** across India:

Subscription Tier	Monthly Fee	Platform Benefits & SLA	Active Resort Allocation
PREMIUM TIER	\$299.00 / mo	Top Priority Discovery, Unlimited Analytics, Full Financial Auditing, 24/7 Support	Taj Falaknuma, Grand Palms, Taj Lake Palace, Wildflower Hall, etc.
SUPER TIER	\$149.00 / mo	Elevated Discovery, Advanced Staff Ops, Standard Reports, Email Support	Highland Mist, Golkonda Resort, Kumarakom Lake, Tamara Coorg, etc.
BASIC TIER	\$49.00 / mo	Standard Discovery, Basic Booking Engine, Core Financial Reporting	Lakeside Eco, Backwater Palms, Valley View Ooty, Emerald Pines, etc.
FREE TIER	\$0.00 / mo	Basic Directory Listing, Manual Confirmation Only	Pine Valley, Himalayan River, Sunset Dunes Camp, Wayanad Coffee, etc.

6.1 Super Admin Core Capabilities

- **Platform Catalog Inspection:** Filter resorts across tiers, search by name or city, and inspect active subscriptions.
- **" + Add New Resort" Onboarding:** Multi-step modal to create new resorts with name, city, state, address, contact coordinates, subscription tier, and room units.
- **Admin Credential Provisioning:** Assign certified Resort Admins to newly listed or unassigned resorts.
- **Financial Governance:** Track SaaS recurring revenue, subscription renewals, and active MRR/ARR metrics.

7. Design System, Theming & Non-Functional Specifications

7.1 Dual-Theme System Specification

Design Token	Light Mode (Luxury Mint)	Dark Mode (Obsidian Forest)	Visual Application
Page Canvas	#EDF6F2 (Soft Mint)	#0D1512 (Deep Obsidian Slate)	Scaffold & background
Card Surface	#FFFFFF (Pure White)	#16221D (Elevated Dark Slate)	Resort cards, dialogs, AppBars
Pills / Chips	#FFFFFF / #EDF6F2	#1E2E28 (Surface Highlight)	Amenity toggles, filter pills
Primary Accent	#10B981 (Emerald Green)	#10B981 (Vibrant Emerald)	Price tags, active filters, icons
Primary Text	#111827 (Dark Charcoal)	#F9FAFB (Crisp Off-White)	Titles, headings, primary copy
Borders	#E5E7EB (Subtle Gray)	#263A32 (Dark Slate Border)	Card outlines, dividers

7.2 Non-Functional & Quality Attributes

- **Theme Reactivity:** Instantaneous app-wide theme switching powered by `ValueListenableBuilder<ThemeMode>` without requiring page reload.
- **PWA Cache Control:** Zero stale file caching with `--pwa-strategy=none` and automated cache busting.
- **Responsive Mobile Frame:** Studio wrapper allowing seamless switching between responsive mobile container and desktop full-width viewports.
- **Accessibility:** WCAG AA compliant contrast ratios across both light and dark color schemes.